

Vitamins & Minerals After Bariatric Surgery

A practical guide to lifelong supplementation

GROUP A

**Sleeve Gastrectomy
& Roux-en-Y Bypass**

Standard postoperative regimen

GROUP B

**SADI & Duodenal
Switch (BPD/DS)**

Expanded malabsorptive regimen

Stephen Klepner, DO · Justin Dhyani, MD

BARIATRIC SURGERY PRACTICE

EDITION 2026.1 · FOR PATIENT USE

SECTION 01

Why every patient needs supplements.

After bariatric surgery, you will not be able to get enough vitamins and minerals from food alone. This is not a matter of will or diet quality — it is a permanent change to how your body absorbs nutrients. Supplements are for life.

What changes after surgery

All bariatric procedures reduce how much food you eat. Procedures that bypass parts of the small intestine — Roux-en-Y, SADI, and Duodenal Switch — also reduce how much your body absorbs from that food. The result is a real risk of deficiency, even in patients who eat well.

The deficiencies we watch for

Thiamine (B1)

Can cause *permanent* neurological injury. Highest risk in the first 6 months and during any prolonged vomiting.

Iron, Folate, B12

Cause anemia — fatigue, weakness, cold intolerance, brittle nails.

Calcium & Vitamin D

Bone loss, osteoporosis, fractures. Damage accumulates silently for years.

Zinc & Copper

Hair loss, skin and nail changes, altered taste, weakened immunity.

Vitamins A, E, K

Higher risk after SADI and DS — fat-soluble vitamins depend on the section of intestine that is bypassed.

Folate — pregnancy

Folate deficiency can cause neural-tube birth defects. Critical for any patient of childbearing potential.

● HOW THIS GUIDE WORKS

Section 02 lists the daily nutrient targets your body actually needs. Sections 03 and 04 give you two ways to meet those targets — a single-product bariatric regimen and a budget-friendly mix-and-match. The final page is a tear-out reference for your fridge.

SECTION 02

What your body needs, every day, for life.

These targets are drawn from the American Society for Metabolic and Bariatric Surgery (ASMBS) nutritional guidelines. They are minimums, not ceilings. Your lab work guides any adjustments.

Nutrient	Sleeve / Roux-en-Y	SADI / Duodenal Switch
Multivitamin w/ minerals	1–2 daily — with iron, folic acid, thiamine	2 daily — bariatric-specific recommended
Thiamine (B1)	≥ 12 mg/day; 50–100 mg preferred	≥ 12 mg/day; 50–100 mg preferred
Vitamin B12	350–500 µg/day oral · or 1,000 µg/month IM	350–500 µg/day oral · or 1,000 µg/month IM
Folic acid	400–800 µg/day (800–1,000 µg if childbearing potential)	400–800 µg/day (800–1,000 µg if childbearing potential)
Elemental iron	18 mg/day · 45–60 mg if menstruating	45–60 mg/day standard
Calcium citrate	1,200–1,500 mg/day, divided	1,800–2,400 mg/day, divided
Vitamin D3	2,000–3,000 IU/day	3,000–5,000 IU/day (often more)
Vitamin A	5,000–10,000 IU/day	10,000 IU/day (water-miscible form)
Vitamin E	15 mg/day	15 mg/day
Vitamin K	90–120 µg/day	300 µg/day
Zinc	8–11 mg/day	16–22 mg/day · pair with copper
Copper	1 mg/day	2 mg/day

! FORM MATTERS

Calcium must be citrate, not carbonate — you no longer make enough stomach acid to absorb the carbonate form. **No gummy multivitamins** — they lack minerals and some B vitamins. **Do not crush enteric-coated pills.**

SECTION 03 · GROUP A

If you had a Sleeve or Roux-en-Y Bypass.

Pick the option that fits your budget. Both options meet the daily targets in Section 02. The simpler option uses one bariatric-specific multivitamin; the budget option mixes drugstore products to reach the same totals.

OPTION 1 · SIMPLER

Bariatric MVI + Calcium Citrate

~ \$35–\$50 per month

WHAT TO BUY

- **1 bariatric chewable MVI** with iron, twice daily (*or one dose if labeled "complete" for bariatric use*)
- **Calcium citrate + D3**, 500–600 mg, twice daily

RECOMMENDED BRANDS

Bariatric Advantage Ultra Solo with Iron · Celebrate Multi with Iron · ProCare Health Bariatric · BariMelts Complete

OPTION 2 · BUDGET

Drugstore mix & match

~ \$15–\$25 per month

WHAT TO BUY

- **Centrum or Flintstones Complete** chewable, *two* per day
- **Calcium citrate + D3**, 500–600 mg, twice daily (Citracal Maximum)
- **Sublingual B12**, 1,000 µg daily
- **Iron 18 mg** if you menstruate (or 45–60 mg ferrous sulfate)

WHY TWO MVIS

A single drugstore MVI does not meet bariatric thiamine and folate targets. Two doses, spaced morning and evening, do.

Your daily schedule

WHEN	TAKE	WITH
Morning	Multivitamin (dose 1) · Sublingual B12 (if separate)	Breakfast + small glass of water with vitamin C
Midday	Calcium citrate · Iron (if needed)	Lunch — at least 2 hr after morning MVI
Afternoon	Multivitamin (dose 2) — only if taking 2 separate MVIs	Snack
Evening	Calcium citrate (dose 2)	Dinner — at least 2 hr after iron

SECTION 04 · GROUP B

If you had a SADI or Duodenal Switch.

SADI and Duodenal Switch bypass more of the small intestine — including the section that absorbs fat-soluble vitamins (A, D, E, K). Your regimen is larger than Group A's, and a standard drugstore MVI is not enough on its own. Lifelong lab work every 6–12 months is non-negotiable.

! WHY THIS SECTION IS DIFFERENT

After SADI/DS you malabsorb fat. Vitamins A, D, E, K are fat-soluble — without supplementation, deficiencies are not a risk, they are *expected*. Night blindness, bone disease, easy bruising, and clotting problems all trace back to skipping this regimen.

OPTION 1 · SIMPLER

DS-Specific Bariatric MVI + Calcium

~ \$55–\$90 per month

WHAT TO BUY

- **DS- or BPD-formulated chewable MVI**, twice daily — contains higher iron, zinc, copper, and water-miscible A/D/E/K
- **Calcium citrate + D3**, 600 mg, three times daily (1,800 mg total)
- **Sublingual B12**, 1,000 µg daily

RECOMMENDED BRANDS

Celebrate Bariatric Three Multi · Bariatric Advantage Advanced EA Multi · ProCare Health Bariatric (DS-formulated) · Bariatric Fusion BPD/DS

OPTION 2 · BUDGET

Drugstore mix & match

~ \$30–\$45 per month

WHAT TO BUY

- **Centrum or Flintstones Complete** chewable, *two* per day
- **Calcium citrate + D3**, 600 mg × 3 daily (Citracal Maximum)
- **Vitamin D3**, additional 2,000 IU daily
- **Vitamin A (water-miscible)**, 10,000 IU daily
- **Vitamin K1**, 300 µg daily
- **Zinc 22 mg + Copper 2 mg** daily (*or zinc-balanced complex*)
- **Iron 45–60 mg** ferrous sulfate, daily
- **Sublingual B12**, 1,000 µg daily

NOTES

The à-la-carte route works but takes more daily attention. For most DS patients we recommend Option 1 once it fits the budget — adherence is the single biggest predictor of long-term outcomes.

Continued on next page → your daily schedule and lab-monitoring requirements.

Your daily schedule

Spacing matters more here than for Group A — iron and calcium compete for the same absorption pathway, and you are taking more of both.

WHEN	TAKE	WITH
Morning	DS multivitamin (dose 1) · B12 sublingual · Iron + vitamin C	Breakfast
Midday	Calcium citrate (dose 1) · Vitamin A (if separate) · Vitamin K (if separate)	Lunch — at least 2 hr after iron
Afternoon	DS multivitamin (dose 2) · Calcium citrate (dose 2)	Snack — at least 2 hr after first calcium
Evening	Calcium citrate (dose 3) · Vitamin D3 (if separate) · Zinc + copper	Dinner

Lab schedule for DS patients

DS patients have the highest deficiency risk of any bariatric procedure. Lab work is not optional — it is how we catch problems before they become symptoms.

! REQUIRED LABS

Frequency: Every 3 months for the first year, then every 6 months for life.

Panel: CBC, iron panel, ferritin, B12, folate, 25-OH vitamin D, calcium + PTH, magnesium, zinc, copper, vitamin A, vitamin E, INR (as proxy for vitamin K), thiamine.

Bring this guide and your supplement bottles to every appointment. If a brand changes or you switch from one option to another, tell us — the doses are not always interchangeable.

SECTION 05

The rules that make your supplements work.

Vitamins and minerals interact. Two pills you took correctly can still leave you deficient if you took them at the wrong time, with the wrong drink, or against the wrong food. These rules are short — memorize them.

Timing

- **Calcium and iron compete** — never take them within 2 hours of each other.
- **Calcium and thyroid compete** — separate by 2 hr. **Iron and thyroid:** 4 hr.
- **Wait 1 hour** between any supplement and any prescription medication.
- **Never more than 500–600 mg of calcium at one time** — your body can't absorb more.

What to take it with

- **Plain water** only. Coffee, tea, and milk interfere with mineral absorption.
- **Iron with vitamin C** — small glass of OJ or a 250 mg tablet. Big absorption boost.
- **Iron on an empty stomach** if you tolerate it; otherwise a small protein snack.
- **All other supplements with food** — better absorption, less nausea.

Forms to avoid

- **No gummy MVIs** — they lack minerals and B vitamins.
- **No calcium carbonate** — citrate only. You no longer have the acid to absorb it.
- **No crushing** of time-release or enteric-coated tablets.
- **No pills larger than your fingertip** for the first month after surgery.

Habits to keep

- **Take everything every day.** A missed week of B12 is fine; a missed month of thiamine is not.
- **Bloodwork on schedule** — every 3 months year 1, then 6–12 months for life.
- **Tell every provider** you had bariatric surgery — especially before any new prescription.
- **If you can't take it or afford it** — call us. Don't just stop.

! CALL YOUR SURGEON IF

Persistent vomiting, confusion, memory problems, numbness or tingling, vision changes, easy bruising, persistent diarrhea, or unexplained weight loss. These can be early signs of thiamine, B12, or fat-soluble vitamin deficiency.

Quick Reference



Sleeve / Roux-en-Y

GROUP A · DAILY

MORNING · WITH BREAKFAST

Bariatric MVI with iron

B12 sublingual 1,000 µg

Vitamin C tablet (*helps iron*)

MIDDAY · WITH LUNCH

Calcium citrate 500–600 mg + D3

AFTERNOON · SECOND MVI DOSE

Bariatric MVI (if taking 2/day)

EVENING · WITH DINNER

Calcium citrate 500–600 mg + D3

SADI / Duodenal Switch

GROUP B · DAILY

MORNING · WITH BREAKFAST

DS multivitamin (dose 1)

Iron 45–60 mg + vitamin C

B12 sublingual 1,000 µg

MIDDAY · WITH LUNCH

Calcium citrate 600 mg + D3

Vitamin A & K (*if separate*)

AFTERNOON · WITH SNACK

DS multivitamin (dose 2)

Calcium citrate 600 mg + D3

EVENING · WITH DINNER

Calcium citrate 600 mg + D3

Extra Vitamin D3 + Zinc + Copper

FIVE RULES THAT DON'T CHANGE

- **Separate calcium and iron by 2 hours.** Separate either from thyroid hormone.
- **Calcium must be citrate, never carbonate.** No more than 600 mg per dose.
- **Iron with vitamin C. All others with food. Plain water only.**
- **No gummies. No carbonate. No straws or large pills for the first month.**
- **Bloodwork on schedule, supplements every day, for life.**